# Supporting Information: Fully Symmetry-Conditioned Rigid-Body Flow Matching for Molecular-Crystal Structure Prediction


Frank Cai*

*Purdue University, West Lafayette, Indiana 47907, United States*

E-mail: frankyc11223@gmail.com


## S1 Rigid-body factorization, gauge, and manifolds

**Intrinsic gauge.** Each crystal is parsed into rigid blocks by a covalent-bond graph with periodic unwrapping; species are keyed by a Weisfeiler–Leman hash and share one reference conformer rotated into a molecule-intrinsic frame (gyration-tensor principal axes, sign-fixed by an element-weighted third moment). Each copy's pose is the Kabsch rotation to that frame, which removes the cross-crystal reference that otherwise floors the absolute orientation $R_m$. The atom positions reconstruct exactly as $\mathrm{cart}_{m,a} = c_m L + R_m \, \mathrm{local}_{m,a}$, so the re-gauge to the relative target $R'_m = R_m R_0^{-1}$ (Eq. 1 of the main text) leaves the rebuilt crystal unchanged.

**Manifolds.** The lattice is carried in a 10-dimensional log-volume/unit-determinant-shape parametrization (prior volume matched to the corpus mean); centroids on torus geodesics with a minimum-image velocity; and orientations on SO(3) geodesics with body-frame angular-velocity target $u_R = \log(R_0^\top R_1)$, advanced by the exponential map. Training uses conditional flow matching[1] with optimal-transport coupling;[2] sampling is RK4 in 50 steps.

**Deployable coset recovery.** For a space group with operations $\{(W_h, t_h)\}$ (pymatgen), each non-reference copy is assigned $\arg\min_h \|(c_m - (W_h c_0 + t_h)) \bmod 1\|$ under the minimum image, giving the generating-operation index; a deterministic $(\mathrm{sg}, h)$ ordering yields shared integer labels (238 total on the diagnostic corpus). This uses only centroids and space-group operations, never the observed rotation, so the label is leak-free and reproducible from a symmetry template at sampling time.

**Training.** Adam, learning rate $3 \times 10^{-4}$, gradient clipping, batch 16; default network width $d$=128 with four attention and four encoder layers; 800 steps for the coset runs, and $K$=4 prior draws for the SO(3)-averaged objective. The template-free predictor reuses the encoder and pair-bias trunk over conformer/centroid/lattice/space-group inputs (no orientation) and is trained to 39.5% held-out top-1 accuracy.

# S2 The $R_{\mathrm{asym}}$ conditional probe

To test whether the gauge-free asymmetric-unit orientation carries any exploitable signal beyond its uniform marginal, a supervised probe was trained to predict $R_{\mathrm{asym}}$ from the lattice, the centroid arrangement, the conformer shape, and the space group. Across three seeds the probe (mean geodesic error 125.2°) does not improve on the best feature-free constant (122.4°) and sits at the no-information level ($\sim -2.4\%$ relative to the constant), confirming that the uniform marginal reflects genuine conditional unpredictability under this conditioning rather than a masked dependence. The claim is therefore operational and scoped: $R_{\mathrm{asym}}$ is not predictable from the conditioning used here (rigid, general-position organic crystals), not that it is unlearnable in principle.

## S3 Crystal-family lattice mask

We parametrize the cell by its metric $G = L^\top L$, which is invariant to the arbitrary orientation of $L$, and take the matrix logarithm $S = \log G$, a symmetric $3\times 3$ matrix with six independent components. Expanding $S$ in an orthonormal symmetric basis $\{B_0, \ldots, B_5\}$ gives coordinates $k_i = \langle S, B_i \rangle$ that separate volume from shape:

$$
\begin{array}{lll}
B_0 = \frac{1}{\sqrt{3}}\, I & k_0 = \frac{1}{\sqrt{3}}\,\mathrm{tr}\, S = \frac{2}{\sqrt{3}} \log V & \text{(volume, always free)} \\
B_1 = \frac{1}{\sqrt{6}}\, \mathrm{diag}(1, 1, -2) & k_1 & (c \text{ vs. } a, b) \\
B_2 = \frac{1}{\sqrt{2}}\, \mathrm{diag}(1, -1, 0) & k_2 & (a \text{ vs. } b) \\
B_3 = \frac{1}{\sqrt{2}}(E_{23} + E_{32}) & k_3 & (\text{off-diag. } {\sim}\alpha) \\
B_4 = \frac{1}{\sqrt{2}}(E_{13} + E_{31}) & k_4 & (\text{off-diag. } {\sim}\beta) \\
B_5 = \frac{1}{\sqrt{2}}(E_{12} + E_{21}) & k_5 & (\text{off-diag. } {\sim}\gamma)
\end{array}
$$

The cell reconstructs as $G = \exp\left(\sum_i k_i B_i\right)$, $L = \mathrm{chol}(G)$; the triangular (canonical-frame) choice is immaterial because the model round-trips the lattice through $k$ and StructureMatcher is frame-invariant. The binary family mask $m(\mathrm{sg})$ frees only the shape coordinates the crystal family permits and freezes the rest to canonical values (Table S1), applied after every sampler step and in the prior. For the hexagonal family, $\gamma = 120°$ makes the $B_5$ component a fixed *nonzero* constant in log-metric space (independent of $a$, since the $a^2$-scaling contributes only to the volume coordinate $k_0$); its value is computed numerically and verified in a unit test. The mask mechanism follows DiffCSP++;[3] our contribution is transporting it into the molecular rigid-body flow and composing it with the orientation coset.

## S4 Finishing and self-conditioning

**Rigid-press finisher.** Each generated crystal is relaxed under a soft Lennard-Jones energy summed over intermolecular atom pairs within a 6 Å neighbor list, with the contact distance

Table S1: Crystal-family lattice mask: free log-metric coordinates per family (space-group ranges); all others are frozen to the canonical value in the last column. The volume coordinate $k_0$ is always free.

| family (SG range) | free $k$-coords | frozen (value) |
|---|---|---|
| triclinic (1–2) | $0, 1, 2, 3, 4, 5$ | — |
| monoclinic (3–15) | $0, 1, 2, 4$ | $k_3{=}0,\ k_5{=}0$ |
| orthorhombic (16–74) | $0, 1, 2$ | $k_3{=}k_4{=}k_5{=}0$ |
| tetragonal (75–142) | $0, 1$ | $k_2{=}k_3{=}k_4{=}k_5{=}0$ |
| trigonal/hexagonal (143–194) | $0, 1$ | $k_2{=}k_3{=}k_4{=}0,\ k_5{=}\mathrm{CANON}_{\mathrm{hex}}$ |
| cubic (195–230) | $0$ | $k_1{=}\ldots{=}k_5{=}0$ |

set to 0.90× the sum of van der Waals radii. The free variables are the asymmetric-unit pose (an SO(3) tangent and a fractional shift per rigid block) and the family-masked cell coordinates $k$; the intramolecular geometry is frozen (rigid body) and every symmetry copy is *regenerated* from the optimized ASU pose via the space-group operations at each step, so the relaxation cannot break the imposed symmetry and never reads the reference structure. Optimization is Adam (learning rate 0.03, $\sim$40–60 steps) with the neighbor list refreshed every 15 steps and a light volume-penalty term ($\lambda_{\mathrm{vol}} = 0.05$) toward the corpus volume per atom.

**Self-conditioning.** With probability 0.5 during training, the model first produces a detached prediction with a null self-conditioning token, Euler-extrapolates it to $t{=}1$ to form a terminal estimate $(\hat{L}, \hat{x}, \hat{R})$, and re-runs conditioned on that estimate. The extrapolation respects each manifold: $\hat{L} = \mathrm{k\_to\_lat}\big(k_t + (1{-}t)\, v_L\big)$, $\hat{x} = \mathrm{wrap}\big(x_t + (1{-}t)\, v_x\big)$ on the torus, and, in the body frame (right multiplication), $\hat{R} = R_t \exp_{\mathrm{SO(3)}}\big((1{-}t)\, v_R\big)$. At sampling time the estimate is carried across the RK4 steps (reusing the final velocity evaluation), and AlphaFold-style *recycling* re-solves the whole trajectory conditioned on the previous pass's output structure. All of these are gated behind configuration flags, leaving the default generation path unchanged.

## S5 Per-seed numbers and tolerance sweep

**Non-reference loss drop, per seed.** Control 29.4/24.4/28.8%; deployable coset 43.2/37.2/42.9%; +SO(3)-averaged 49.2/43.4/50.5%. At the 2,539-crystal scale corpus: coset 53.5/54.4/55.1%; control 35.2/36.0/39.1%.

**Orientation-isolated reconstruction** (coset-conditioned, best-of-8), StructureMatcher (length, site) tolerance sweep: $(0.2, 0.2) \to 4.6\%$; $(0.3, 0.5) \to 27.5\%$; $(0.5, 0.7) \to 37.4\%$; oracle 100% throughout; median non-reference orientation error 41.1°.

**Template-free.** Predicted-coset (top-1) reconstruction: 0% match, 64.8° error, versus 41.1° template-supplied; top-5 marginalization 51.2°.

**End-to-end, raw flow** (all manifolds sampled, best-of-20, no finisher): 0% match; median errors lattice-parameter 0.44, centroid 0.35, orientation 13.4°.

**End-to-end, fully symmetry-conditioned + finished** (main-text Table 7, strict match@10): best 6.9% at stol 1.0 (95% CI 3.66–12.54) / 6.1% at 0.8; median atom-RMSD $1.02 \to 0.77$, cell-volume RMAD $33\% \to 21\%$, cell angles within 2° of 90° $16\% \to 73.6\%$ (reference 74.0%).

## References

(1) Lipman, Y.; Chen, R. T. Q.; Ben-Hamu, H.; Nickel, M.; Le, M. Flow Matching for Generative Modeling. International Conference on Learning Representations (ICLR). 2023; arXiv:2210.02747.

(2) Pooladian, A.-A.; Ben-Hamu, H.; Domingo-Enrich, C.; Amos, B.; Lipman, Y.; Chen, R. T. Q. Multisample Flow Matching: Straightening Flows with Minibatch Couplings. International Conference on Machine Learning (ICML). 2023; arXiv:2304.14772.

(3) Jiao, R.; Huang, W.; Liu, Y.; Zhao, D.; Liu, Y. Space Group Constrained Crys-

tal Generation. International Conference on Learning Representations (ICLR). 2024; arXiv:2402.03992; the DiffCSP++ model.